STOCK MARKET DATA PIPELINE

REVIEW – 1

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Abstract

- The **Stock Market Data Pipeline** is a data engineering project designed to collect, process, transform, store, and analyze stock market data efficiently. Stock market information such as stock prices, trading volume, opening price, closing price, highest price, and lowest price is generated continuously from financial data sources.
- The proposed pipeline automatically collects this data using **Python**, cleans and transforms the raw data, and stores it in a structured database or data warehouse. The processed data can then be used for analysis, visualization, and generating useful insights about stock market performance.
- The project demonstrates important data engineering concepts such as **data ingestion, data cleaning, ETL/ELT processing, data transformation, data storage, pipeline automation, and data visualization**. The system can help users monitor historical and current stock information and provide reliable data for further financial analysis.

Introduction

- The stock market generates a huge amount of data every day. Stock exchanges continuously produce information about companies, prices, trading volumes, market movements, and transactions.
- Raw stock market data may come from different sources and may contain missing values, duplicate records, incorrect formats, or inconsistent timestamps. Therefore, simply collecting the data is not enough. The data must be properly processed before it can be used for analysis.
- A **data pipeline** provides a systematic way to move data from its source to a destination where it can be analyzed.
- The proposed **Stock Market Data Pipeline** collects stock data from an external financial API or dataset, processes the raw information using Python, performs data cleaning and transformation, and stores the processed information in a database.
- The pipeline can then provide data to visualization tools such as **Power BI, Tableau, or Python dashboards**.
- **Basic Pipeline**
- **Data Source → Data Ingestion → Data Cleaning → Data Transformation → Data Storage → Data Analysis → Visualization**
- This approach demonstrates how data engineering techniques can be applied to a real-world financial data problem.